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DIV	BE COMPS ADV Batch A
ADV EXP 2	

AIM	Create advanced charts using Tableau / Power BI / R / Python / D3.js to be performed on the dataset - Socio economic data • Advanced - Word chart, Box and whisker plot, Violin plot, Regression plot (linear and nonlinear), 3D chart, Jitter • Write observations from each chart
Dataset Particulars	Name: Big Basket Dataset Link: https://www.kaggle.com/datasets/nelgiriyeewithana/countries-of-the-world-2023 Dataset Features : 1) Country: Name of the country. 2) Density (P/Km2): Population density measured in persons per square kilometer. 3) Abbreviation: Abbreviation or code representing the country. 4) Agricultural Land (%): Percentage of land area used for agricultural purposes. 5) Land Area (Km2): Total land area of the country in square kilometers. 6) Armed Forces Size: Size of the armed forces in the country. 7) Birth Rate: Number of births per 1,000 population per year. 8) Calling Code: International calling code for the country. 9) Capital/Major City: Name of the capital or major city. 10) CO2 Emissions: Carbon dioxide emissions in tons. 11) CPI: Consumer Price Index, a measure of inflation and purchasing power. 12) CPI Change (%): Percentage change in the Consumer Price Index compared to the previous year. 13) Currency_Code: Currency code used in the country. 14) Fertility Rate: Average number of children born to a woman during her lifetime.

	15) Forested Area (%): Percentage of land area covered by forests. 16) Gasoline_Price: Price of gasoline per liter in local currency. 17) GDP: Gross Domestic Product, the total value of goods and services produced in the country. 18) Gross Primary Education Enrollment (%): Gross enrollment ratio for primary education. 19) Gross Tertiary Education Enrollment (%): Gross enrollment ratio for tertiary education. 20) Infant Mortality: Number of deaths per 1,000 live births before reaching one year of age. 21) Largest City: Name of the country's largest city. 22) Life Expectancy: Average number of years a newborn is expected to live. 23) Maternal Mortality Ratio: Number of maternal deaths per 100,000 live births. 24) Minimum Wage: Minimum wage level in local currency. 25) Official Language: Official language(s) spoken in the country. 26) Out of Pocket Health Expenditure (%): Percentage of total health expenditure paid out-of-pocket by individuals. 27) Physicians per Thousand: Number of physicians per thousand people. 28) Population: Total population of the country. 29) Population: Labor Force Participation (%): Percentage of the population that is part of the labor force. 30) Tax Revenue (%): Tax revenue as a percentage of GDP. 31) Total Tax Rate: Overall tax burden as a percentage of commercial profits. 32) Unemployment Rate: Percentage of the labor force that is unemployed. 33) Urban Population: Percentage of the population living in urban areas. 34) Latitude: Latitude coordinate of the country's location. 35) Longitude: Longitude coordinate of the country's location. 36) The shape of the dataset is : [195* 35]
Data Visualisations	WORD CLOUD



Fig 1.

1. As seen above the highest word count is seen of "English" followed by "French" Countries.
2. This suggests that most countries have English as their Official language

BOX AND WHISKER PLOT

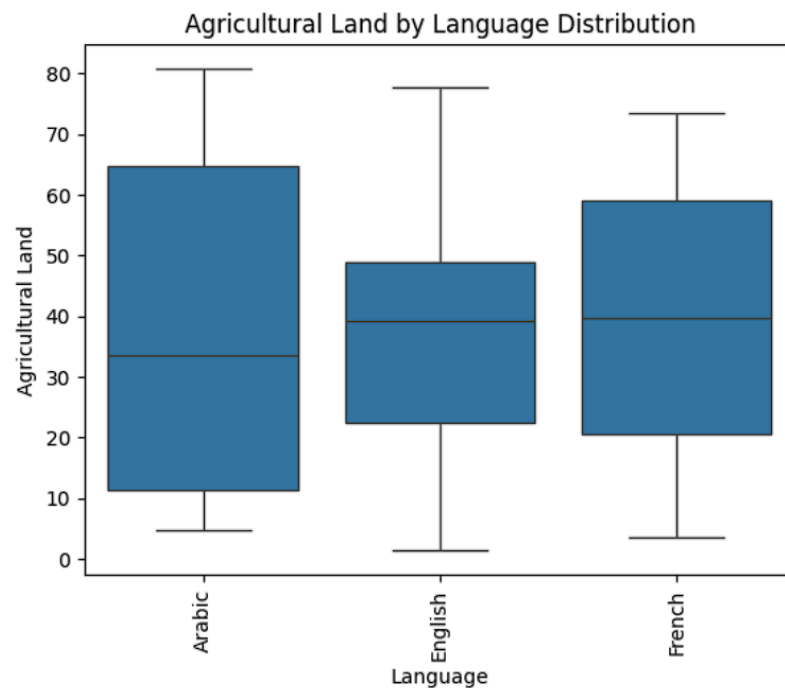


Fig 2.

English Countries have a relatively even distribution of Agricultural Land, where as Arabic countries have a significant variance in agricultural land

Countries with French as their official language have the highest median Agricultural land while Arabic have the minimum median Agricultural land.

REGRESSION CHART - LINEAR

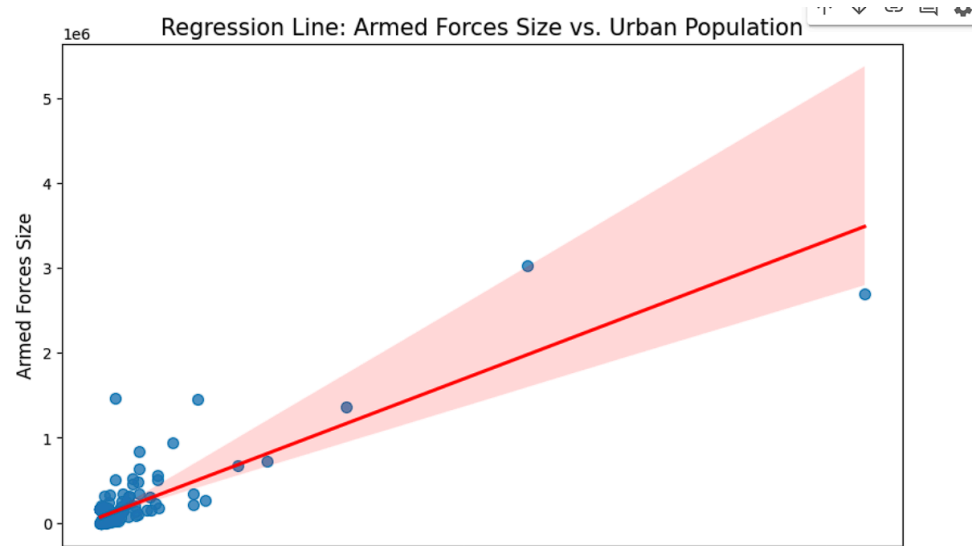


Fig 3

The Armed Forces Size of any country is directly proportional to the Urban Population size and doesn't follow a linear trend.

NON-LINEAR REGRESSION

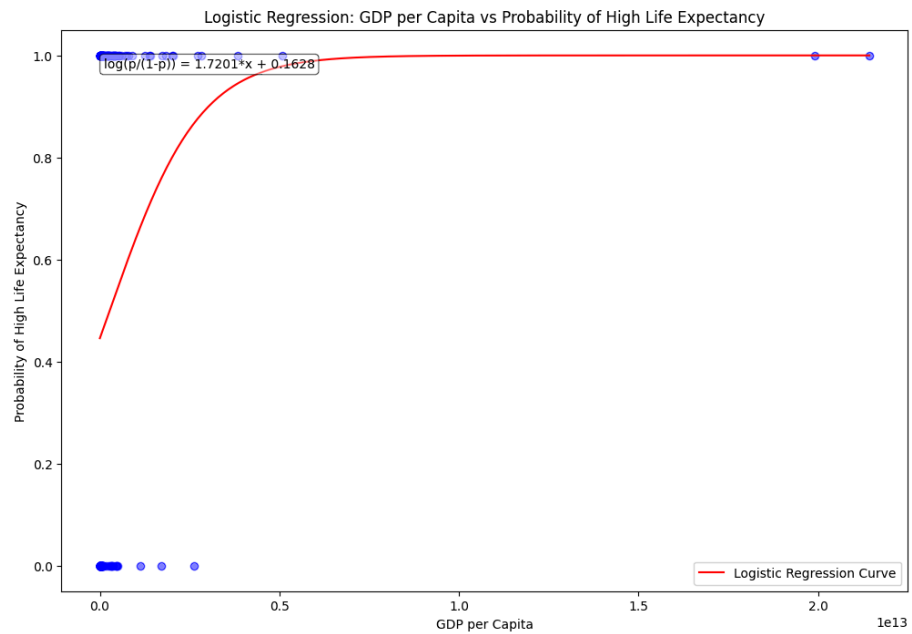


Fig 4

Countries with a higher GDP per capita are more probable to have a higher life expectancy.

Not all countries with low a GDP have got low -life expectancy

3D SCATTER PLOT

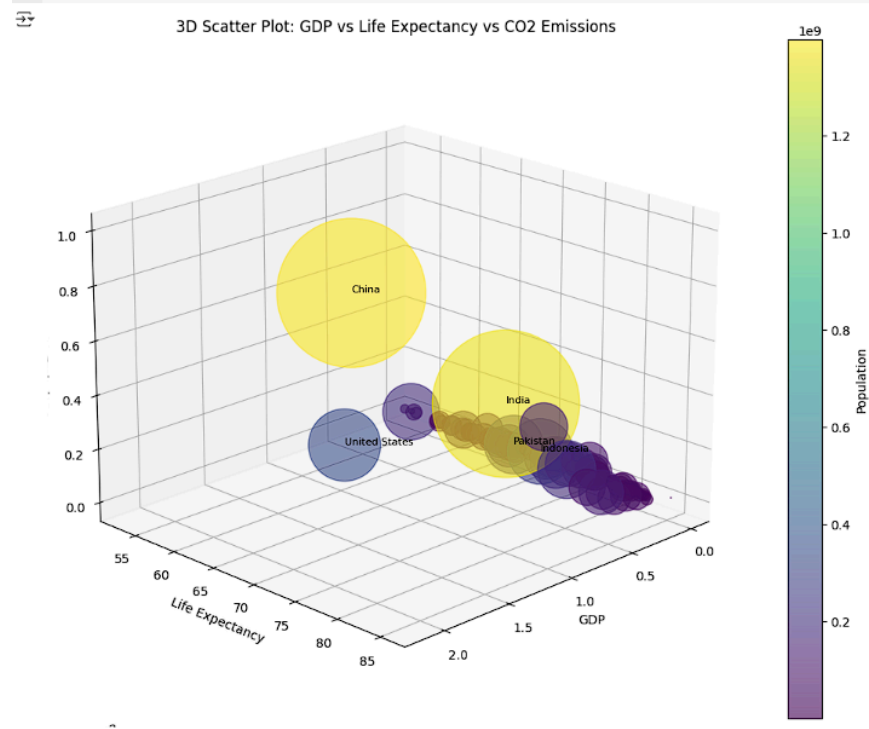


Fig 5

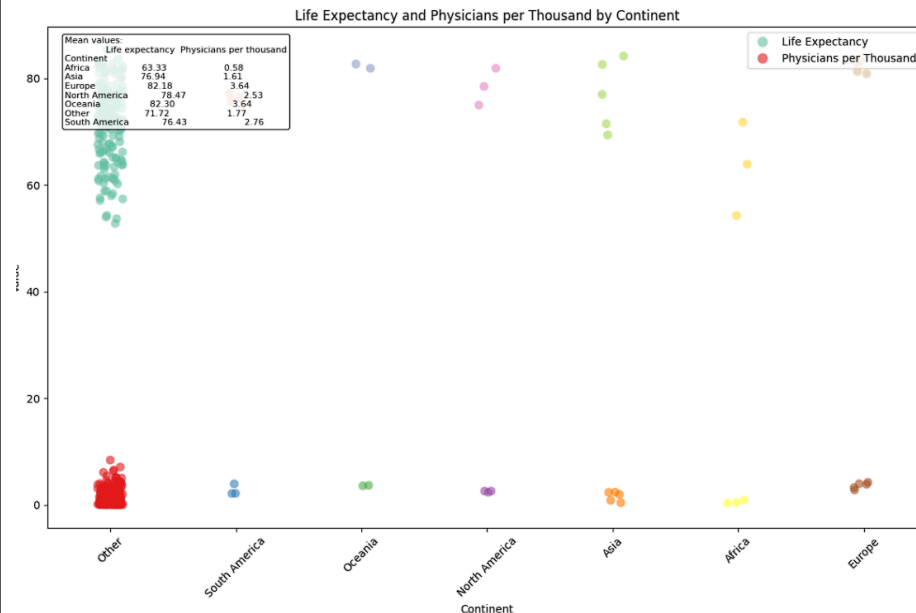
Shows the relationship between 3 features - GDP, Life Expectancy and CO2 emissions

GDP, life expectancy, and CO2 emissions generally increase together.

China and India stand out with high emissions and populations.

Developed nations cluster with high GDP and life expectancy.

JITTER PLOT



Life expectancy data clusters between 60-85 years, while physicians per thousand ranges from 0-5.

Africa shows the lowest values for both metrics, with Europe at the opposite end, having the highest.

Other continents fall between these extremes, with varying distributions of life expectancy and physician density.

Suggestions

- 1) Reduce SKUs and focus on the 5 best selling ones
- 2) Keep higher stock of Sets, Kurtas and lower stocks of the rest
- 3) Stocks in each state should be calculated on the basis of ratio of each category sold in that state
- 4) XS,S,M,L,XL,XXL should have stocked up inventory, all other sizes can be inventory-light and custom made
- 5) Focus on expanding in Maharashtra, Karnataka and drive up average order value
- 6) Take measures for customer retention

	7) Push for offers during holiday season to drive up sales
Conclusion	By studying the visualizations in the dataset above, I was able to analyze the trends in Fashion on Amazon and come up with suggestions for improving service and revenue.